

Out of Sight, but Not Out of Mind

BLINDSIGHT. Research has described several cases of blind people who nevertheless can perceive objects and colors around them. The unusual phenomenon is at the center of a major dispute about consciousness.

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If you do not know the story behind it, the 39-second video from 2008 is fairly unimpressive. It shows an elderly man in an unbuttoned shirt carefully walking down a hospital corridor, followed closely by a researcher with a broad grin on his face.

On the short walk, the man makes his way around a series of obstacles placed on the floor. First a trash can, then a camera tripod; then he squeezes between a stack of paper and a paper tray before edging around a box. What is remarkable is that the man is blind. But not in any ordinary sense.

In neuroscience, the man is known by his initials, T.N., because he has appeared in numerous scientific papers. T.N. was one of the few people in the world who could visually register things around him without being able to see. Yes, that sounds thoroughly contradictory, and so does the name of the phenomenon: blindsight.

Blindsight challenges the common idea that conscious experience is necessary in order to perceive visual input and act rationally on it. Since the phenomenon was discovered in humans in the 1970s, many researchers have used it as evidence for theories about the nature of consciousness. Some of these theories argue against the idea that consciousness is a prerequisite for intelligent behavior. If the brain can recognize facial expressions and control complex movements without any conscious visual experience, what purpose does consciousness serve?

A recent research paper suggests, however, that T.N. actually did have a faint conscious visual experience. And that is grist to the mill for researchers who have criticized work in the field for drawing hasty conclusions.

“In consciousness research, blindsight has long been cited as an important case showing that one theory or another is correct. My own view is that advocates of those theories are no longer entitled to use blindsight in that way,” says Ian Phillips, professor of philosophy and psychology at Johns Hopkins University.

THE FEELING OF SEEING

In a 2025 paper, researchers revisited old video recordings of their conversations with the patient and discovered that they had overlooked important details. One of the most striking findings when they reviewed the recordings was T.N.’s ability

to point out colored objects and describe how color vision “felt,” according to the paper.

In consciousness research, it has long been said that conscious experiences have a distinctive phenomenal character. That is, there is something it feels like to have a subjective experience, whether it is a shooting pain in the lower back, a bitter taste in the mouth, or the breathtaking sight of a sunrise. Understood in this way, T.N.’s remarks are worth paying attention to.

“Originally, they presented the patient as being completely blind and unaware of how he made his way through the obstacle course. But in the transcripts you can read that T.N. repeatedly indicates that he can see things, including colors,” says Ian Phillips.

For example, T.N. is shown a red A4-sized brochure from a distance of two meters, which initially gives him trouble. But when the researcher moves closer, he suddenly exclaims: “Wait, wait, wait, wait ... yes, the color is coming through, but ... red hasn’t quite come through yet, but ...” He continues looking at the brochure for 10 to 15 seconds before saying: “Yes, there it is, I can see the red color now ... it’s red.”

Ian Phillips believes that if someone can report seeing colors, there must be a conscious experience present - even if it is a weak visual experience unlike ordinary vision. He has long argued that many experiments with blindsight patients have methodological weaknesses. Often they are shown figures on a computer screen in the blind region of the visual field and asked to report whether they can see the figure or not. The problem is that nuances can easily be lost with this method. “Binary questions force you to fit the experience into one of two possible categories. There is therefore reason to think that people with weak or degraded visual experiences sometimes answer that they have no visual experience,” he says.

If, instead, blindsight patients are tested with a graded scale, it becomes clear that there are more shades to their answers than they can express when the question is simply either-or.

In a 2008 study, a group of researchers including scientists from Aarhus University used just such a graded scale when testing a 31-year-old woman known as G.R. After a hemorrhage on the left side of the occipital lobe, she lost vision in the upper-right quadrant of her visual field, but she was still able to identify objects in that area. In other words, she had blindsight.

First, she was asked to guess which figures were being shown on a screen in her blind region. She was also asked to report whether she could see the figures or not. As expected, she was able to identify many of the figures even though she said she could not see them.

In the next experiment, she performed exactly the same task, but this time she had to classify her experiences into four categories: clear image, almost clear image, faint glimpse, or no vision. She told the researchers that these categories corresponded better to her experiences.

The results showed a clear relationship between the vividness of her visual experience and the accuracy of her guesses. The researchers therefore concluded that "G.R.'s ability to discriminate between visual stimuli does not reflect unconscious vision, but rather degraded, yet conscious vision."

TWO TYPES OF CONSCIOUSNESS?

On the other side of the debate is Nicholas Humphrey, professor emeritus of psychology at the London School of Economics.

"Blindsight is an absolutely crucial phenomenon for understanding consciousness."

NICHOLAS HUMPHREY

Back in 1964 he began his PhD in psychology, studying monkeys' visual perception and the underlying mechanisms in the brain. It was in this context that he met the rhesus monkey Helen, who later became a scientific celebrity because of her remarkable abilities.

Helen had had the visual cortex removed from her brain, leaving her blind. The impairment had made her timid, and she would sit by herself in a corner of her cage.

But there were small signs that, despite the operation, she could distinguish light from darkness. The young Humphrey became interested in her and began investigating whether there was more to it.

So when his supervisor went abroad to a conference, Humphrey began carrying out a series of tests. He held up a piece of apple in front of Helen, and she reached for it. Humphrey then did the same with other objects, and again and again the monkey reached out for them.

In a burst of excitement, Humphrey sent his supervisor a telegram: "You won't believe this. I've taught Helen to see."

Sure enough, his supervisor did not believe him. But he soon had to concede that his student was right. A month later, the work resulted in the first scientific paper on the subject, published in the prestigious journal *Nature*.

Among consciousness researchers, the hope has been that the phenomenon could help explain how the brain contributes to conscious experience.

Nicholas Humphrey believes that blindsight reveals two different kinds of consciousness, which he calls "cognitive consciousness" and "phenomenal consciousness."

The first contains neutral information about an object's size, shape, location, and movement without providing any subjective experience of what the object looks like. The second is sensory and contains qualitative aspects of objects that give rise to particular subjective impressions.

According to Humphrey, blindsight patients lack phenomenal consciousness while their cognitive consciousness remains intact.

The challenge for Humphrey, however, is to explain how a patient such as T.N. can see colors, which are generally regarded as sensory.

"T.N. says that he does not see colors in the way he used to see them. Other patients with blindsight have also been able to name colors in their surroundings, even though they say that they do not have color-related sensory impressions," says Humphrey.

"The reason I do not think the patient has sensory experiences is that he himself says so."

BACK TO THE DRAWING BOARD

The disagreement among the researchers illustrates that there are different views about what consciousness is and how it should be measured.

"Although subjective reports should be taken seriously, they should not always be taken at face value," says Ian Phillips.

His challenge to the idea of unconscious visual abilities has, in fact, been based on taking a critical approach to patients' statements. Over time, more researchers have had to concede that better methods have shown that in some cases blindsight is simply impaired vision - and therefore not unconscious. As a result, the number of cases supporting the idea of unconscious vision has narrowed.

"People interested in blindsight rely a little too often on superficial treatments of the topic and take them at face value. But there is now so much literature on the other side of the debate pointing out the problems that it is difficult to overlook how controversial the subject is," he says.

In light of these controversies, researchers need to go back to the drawing board, he says.

"The positive side of the matter is that it clears the way for the traditional view that one has to be conscious of something before one can act intentionally on the basis of it. I am inclined to think that this is a powerful idea that theories of consciousness should take seriously," says Ian Phillips.

"It tells us why consciousness is valuable: it helps us act in the world."